



Tone on a Budget: A Reference-Free Metric for Lexical Tone in Massively-Multilingual TTS

Yorùbá tone is near-native *zero-shot* — and CER can't tell you that

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The problem

African tonal languages are underserved by speech synthesis. New *massively-multilingual* TTS models speak Yorùbá *zero-shot*, but a leading one (OmniVoice, 646 languages; arXiv:2604.00688) posts its worst zero-shot CER on Yorùbá among the five African languages OmniVoice reports (21.4%, worse than its own ground-truth audio). Two questions block progress:

- Q1 — **Data**: how much correct *lexical tone* is already present *zero-shot*, and what does a few hours of clean target-language audio actually buy?
- Q2 — **Measurement**: how do we even *score* tone fidelity? The standard metric, **Character Error Rate (CER)**, is **blind to tone**.

Why CER cannot see tone

Yorùbá has **3 contrastive tones** (High / Mid / Low). They change *words*, not just melody:
ọkọ (husband) $\neq$ **òkò** (vehicle) $\neq$ **ókó** (hoe)
A synthesiser can be *perfectly intelligible* (low CER) yet say the *wrong word* by mis-tuning pitch. Intelligibility $\neq$ tone.
We need a tone-specific, reference-free metric — one that needs no parallel ground-truth recording of the test sentence.

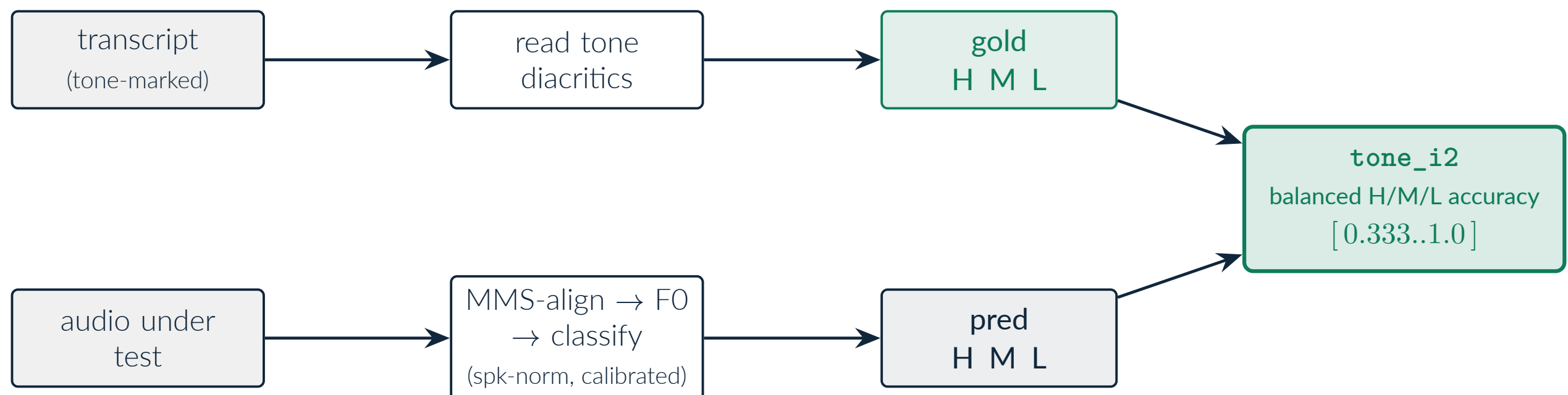
Method 1 — a stable low-resource TTS substrate

We finetune OmniVoice [1] (k2-fsa): an open (Apache-2.0), massively-multilingual **non-autoregressive, masked-generative** TTS (Qwen3-0.6B backbone, frozen Higgs-Audio-v2 codec [3], 24 kHz) — chosen because it already speaks Yorùbá zero-shot and is openly finetunable.

- Clean, single-speaker, *tone-marked* Yorùbá (BibleTTS [13] + OpenSLR-86 [14]), **28.5 h**; $1r \times 10^{-5}$.
- Stable where AR collapses**: in our setup an autoregressive TTS (Qwen3-TTS [2]) *collapsed* at the learning rates needed to install tone; the masked-generative decoder did not. (Single comparison — backbone, codec and training also differ.)

Method 2 — a reference-free tone metric (tone_i2)

A pitch-based meter that scores *tone* directly. **tone_i2** = balanced H/M/L tone-classification accuracy in [0.333 (chance)..1.0]; higher = better.



✓ **No reference recording needed**. The gold tone sequence (top row) is read straight from the transcript's diacritics; only the audio path (bottom row) touches the signal — MMS [10] alignment, then F0 classification into pred. **Sanity-checked** for pitch-sensitivity against a PSOLA [4] pitch-flattening *oracle* (does the score fall when we destroy tone but keep words? — yes, see Result 2), which also proves **CER is tone-blind**. Native-speaker pilot (1 Yorùbá listener, 24 controlled A/B tone pairs): correctly-toned clip picked **96%** ($p < 0.001$ vs. chance); **tone_i2** agrees at **AUROC 0.96** — first human evidence the metric tracks perceived tone ($N=1$ — point estimate only, CI uninformative; preliminary).

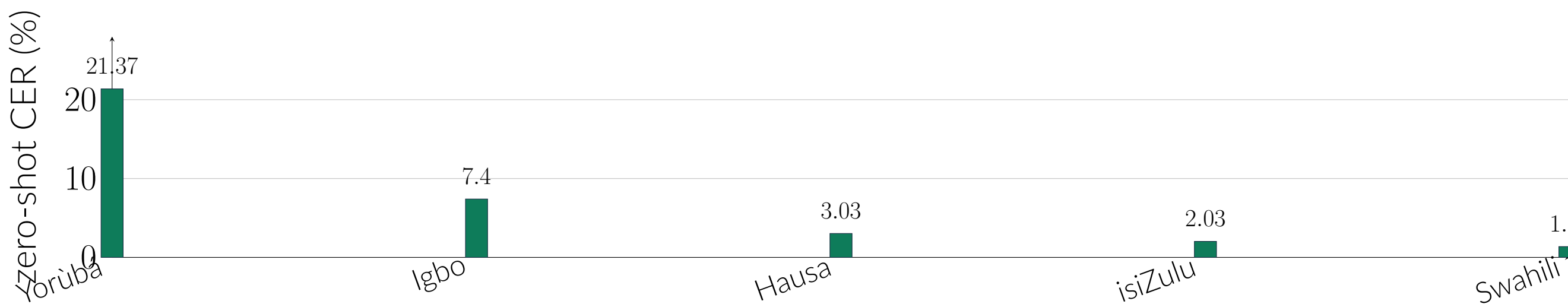
Vs. *prior art*: reference-free TTS/prosody metrics exist (TTSScore [5], prosodic-ABX [6]); tone has been probed in discrete units [7]. Prior **tone-error-rate (TER)** metrics are either *reference-based* (need ground-truth tone labels; Yorùbá / African ASR [8],[15]) or *human-judged* (TTS [16]); **tone_i2** is the first *automated, reference-free* TER for *synthesized* African speech — per tone-bearing unit (TBU), oracle-validated (toolkit also bundles an AfriHuBERT [9] probe).

Quality is preserved, not traded

Tone gains do *not* cost intelligibility or voice identity:
CER **5.6%→1.8%** · voice-clone SSIM $\approx$ **0.83** (flat across budgets)
Intelligibility improves and voice identity holds steady as tone improves. (SSIM = automatic ECAPA [12] cosine to held-out speakers; a human voice-identity check is pending.)

Toward a 6-language African tone benchmark

Why Yorùbá first? Of our African targets, OmniVoice's zero-shot intelligibility is *worst* for Yorùbá — yet CER calls the rest “easy” while being unable to see whether the *tonal* ones get tone right:



Paper-reported OmniVoice zero-shot CER (k2-fsa, arXiv:2604.00688) — its own benchmark/ASR, not comparable to our held-out, in-domain CER. Tonal: Yorùbá, Igbo, Hausa, isiZulu; Swahili non-tonal.

Roadmap: Yorùbá ✓ (tone) · Swahili ✓ (generalizes, non-tonal) · Hausa next (verified clean data, 86 h) · Igbo tone-benchmark from a tone-marked dictionary · isiZulu needs new recordings · Nigerian English (non-tonal) — all with **native-speaker ABX**.

Generalization beyond Yorùbá — Swahili (non-tonal)

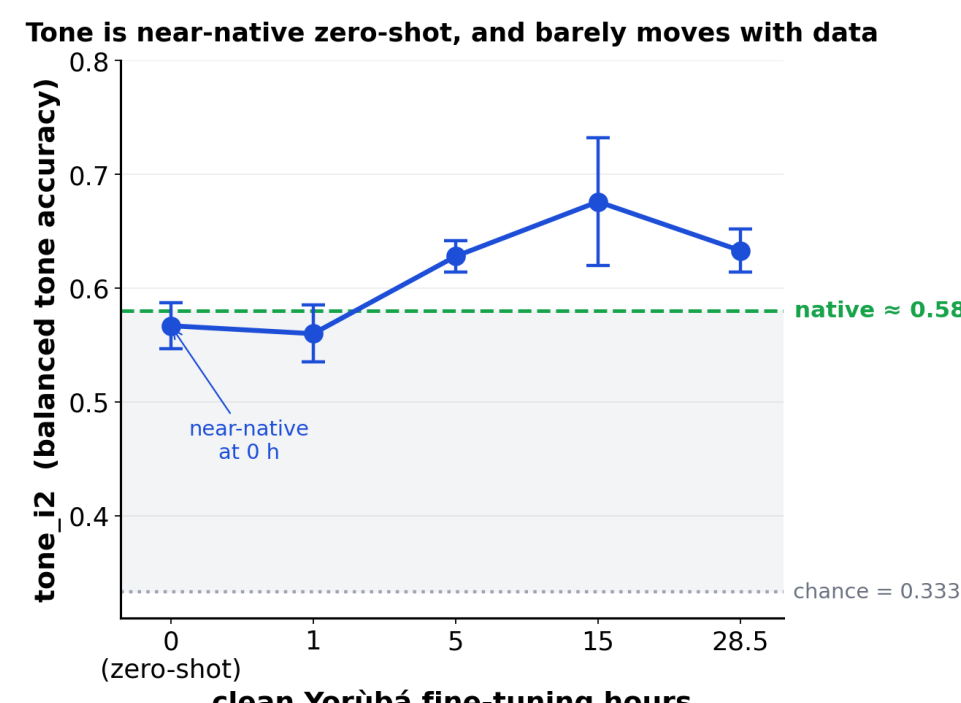
We repeat the data-efficiency sweep on Swahili — a *non-tonal* language — finetuning OmniVoice on clean, single-speaker OpenBible audio.

hours	CER	SSIM
0 (zero-shot)	0.055	0.886
1	0.036	0.888
2.5	0.042	0.875
5	0.029	0.879

All Swahili budgets single-pass (no error bars); 1–5 h within run-to-run noise. CER = Whisper-large-v3 [11] on held-out OpenBible.

Same finding, mirror image: 1 h of in-domain audio already cuts CER $\sim$ 35% (0.055→0.036), reaching $\sim$ 47% by 5 h (0.029); voice fidelity holds (SSIM $\approx$ 0.88). **For a non-tonal language CER is enough** — there is no hidden tone axis. The protocol transfers: the *metric you need* depends on the language.

Result 1 — near-native tone zero-shot; a few hours → 3× fewer ASR errors



tone_i2 vs. clean Yorùbá hours (mean±std, 3 training seeds per budget).

hours	tone_i2 (mean±std)	CER	SSIM
0 (zero-shot)	0.567 ± 0.020	0.056	0.82
1	0.560 ± 0.025	0.036	0.83
5	0.628 ± 0.014	0.018	0.84
15	0.676 ± 0.056	0.021	0.83
28.5	0.633 ± 0.019	0.022	0.83

native $\approx$ 0.60 (0.596, 300 clips/37 spk) · chance = 0.333 · 1/5/15 h bars: **3 training seeds**; 0/28.5 h bars: **decode-seed spread** (single run) · CER: our held-out ASR.

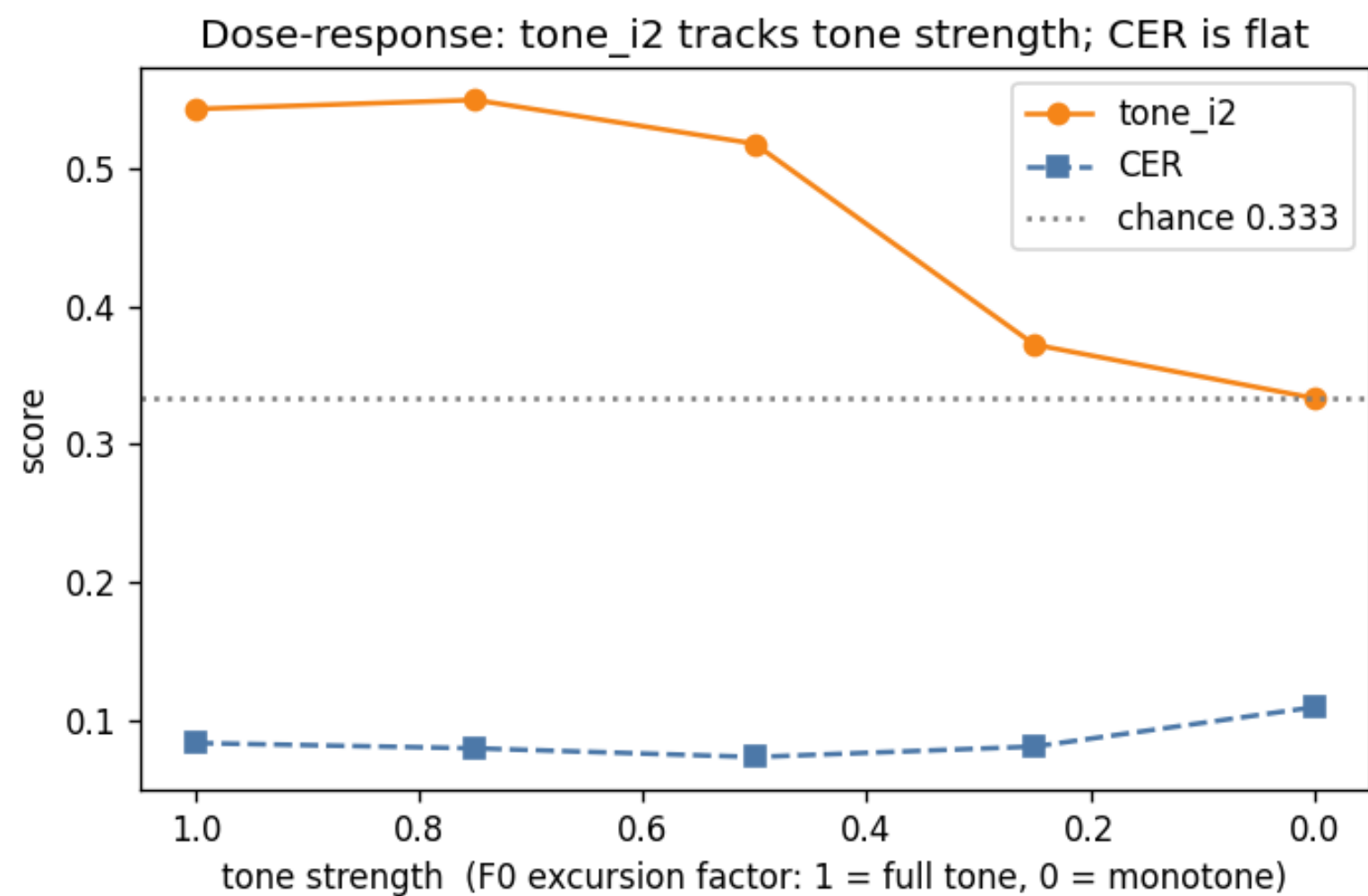
- Near-native tone zero-shot** (tone_i2 0.57 vs native 0.60) — and 1 h is not yet distinguishable from it (0.56).
- What a few hours buys is intelligibility, not tone**: CER drops 5.6%→1.8% ($\sim$ 3×) by 5 h, while tone_i2 only nudges 0.57→0.63 (a small move on a noisy axis); voice identity flat (SSIM 0.83).
- 5 h $\approx$ 15 h within training variance** (3 seeds; 15 h is high-variance — one seed 0.74, two at 0.64): the gain is real by 5 h but does *not* reliably grow to 15 h. (Native reference $\approx$ 0.60; **tone_i2** saturates near native — systems at/above it aren't reliably ranked on this noisy axis.)

Result 2 — oracle test: the meter sees tone, CER is blind

(a) **Controlled**: progressively *flatten* pitch (PSOLA, scale 1.0→0.0). **tone_i2 falls monotonically** (corr +0.92) while **CER stays flat** (0.07–0.11).

(b) **In the wild**: over the real data sweep the two move at *different rates* — **0→5 h CER falls $\sim$ 3× (0.056→0.018) while tone_i2 barely shifts (0.57→0.63, within noise)**; 5→15 h neither separates. Intelligibility and tone are not the same axis — a distinction **CER alone cannot make**.

(c) **Flip the answer key**: on 300 held-out native clips (37 speakers), scoring correct audio against tone-scrambled transcripts collapses **tone_i2 0.60→0.34** (chance); a High↔Low swap drives the 2-class readout to **0.14 — below its 0.50 chance**, an anti-correlation a tone-blind score cannot produce. Predictions held byte-identical; only the gold changed (paired $\Delta = 0.28$, 95% CI [0.26, 0.30]).



Flattening pitch drops **tone_i2** while CER is unmoved.

Takeaways

- A reference-free, lexical-tone metric (tone_i2)** that catches degradations *CER is structurally blind to* — oracle-validated two ways (pitch-flattening and answer-key-flip controls). **This is the contribution**.
- Yorùbá tone is already near-native zero-shot** — a few hours of clean audio buy *intelligibility* (CER 5.6%→1.8%), *not* tone, at no cost to voice identity. **CER alone would tell you the wrong story**.
- The metric you need is language-dependent**: the recipe transfers to a non-tonal language (Swahili), where CER alone suffices — no tone axis to miss.

Limitations: **tone_i2** is validated on *one* tonal language (Yorùbá) — Igbo downstep, Hausa & isiZulu tone systems may need re-calibration; human evidence is $N=1$ (preliminary).

Hear what CER misses



Scan, then open the **Listen** section of the README to A/B zero-shot vs. finetuned saying **ókó** (hoe) vs. **òkò** (vehicle), plus a clip whose words stay perfect while the *tune* is destroyed — low CER, wrong word.

References

- [1] H. Zhu et al., “OmniVoice: Towards Omnilingual Zero-Shot TTS with Diffusion Language Models,” arXiv:2604.00688, 2026. [2] Qwen Team, “Qwen3-TTS Technical Report,” arXiv:2601.15621, 2026. [3] Boson AI, “Higgs Audio V2 Tokenizer,” HuggingFace, 2025. [4] E. Moulines, F. Charpentier, “Pitch-synchronous waveform processing for TTS using diphones,” Speech Comm. 9(5–6):453–467, 1990. [5] I. R. Ulgen et al., “Objective Evaluation of Prosody and Intelligibility in Speech Synthesis (TTSScore),” arXiv:2509.20485, 2025. [6] H. Sun et al., “Prosodic ABX: Measuring Prosodic Contrast in Speech Representations,” arXiv:2604.02102, 2026. [7] O. Osakuade, S. King, “Lexical Tone is Hard to Quantize: Probing Discrete Speech Units in Mandarin and Yorùbá,” arXiv:2604.07467, 2026. [8] S. G. B. Bengono Ojiambo et al., “Improving Tone Recognition using Wav2Vec2 Representations in Yorùbá,” ACM TALLIP 23(12), 2024. [9] J. O. Alabi et al., “AfriHuBERT: A self-supervised speech model for African languages,” arXiv:2409.20201 [Interspeech 2025]. [10] V. Pratap et al., “Scaling Speech Technology to 1,000+ Languages (MMS),” arXiv:2305.13516, 2023. [11] A. Radford et al., “Robust Speech Recognition via Large-Scale Weak Supervision (Whisper),” arXiv:2212.04356, 2022. [12] B. Desplantes et al., “ECAPA-TDNN,” Interspeech 2020, 3830–3834. [13] J. Meyer et al., “BibleTTS: a large, high-fidelity, multilingual African speech corpus,” Interspeech 2022 (OpenSLR SLR129). [14] A. Gutkin et al., “Developing an Open-Source Corpus of Yorùbá Speech,” Interspeech 2020, 404–408 (OpenSLR SLR86). [15] F.-Y. Chen, L. Adeleke, C. M. Downey, “Linguistically Informed Evaluation of Multilingual ASR for African Languages,” AfricaNLP 2026, 149–162 (arXiv:2602.04716; defines a reference-based Tone Error Rate). [16] A. Mohanta, Remuatpui, P. Sarmah, R. Sinha, W. Lalminghlui, “Towards Prosodically Informed Mizo TTS without Explicit Tone Markings,” arXiv:2601.02073, 2026 (human-judged TER over tone-bearing units).

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